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Analysis of Web-based Systems

Project nr 3

Authors of the project:

Jakub Kamiński, Adam Ćwikliński

Wrocław University of Science and Technology

Faculty of Computer Science and Telecommunications

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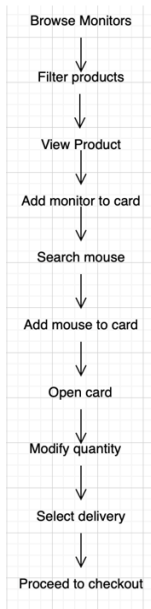
Agenda

- ➊ Introduction
- ➋ Architecture of the online shopping process
- ➌ Performance measurement of websites
- ➍ E2E testing
- ➎ Conclusions



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Introduction





Part 1: E-commerce Architecture Analysis

Objective: Simulate and analyze a complete shopping process on the Novatech platform using Chrome DevTools and Proxyman.

- **Key Actions:** Browsing monitors, adding to cart, searching for a mouse, modifying quantities.
- **Communication:** Browser dynamically requests `/includes/minibasket.html` via AJAX after every action.
- **Result:** UI is refreshed asynchronously without reloading the entire page.

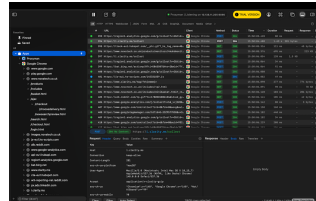


Fig 1. Dev Tools network tab analysis.



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Part 1: Security Verification

Can we manipulate the product price using browser tools?

The Test Scenario:

- Monitor added to cart
(Original price: **£229.99**).
- Client-side DOM
manipulation via DevTools
(Modified to: **£179.99**).

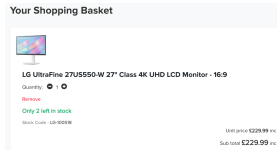


Fig 2. Actual price.



Fig 3. Price after change.

Outcome & Conclusion

The change was strictly visual. Upon refreshing or proceeding to checkout, the server restored the original price.

Proof: Price validation is securely handled on the backend.



Part 2: Performance Indicators (Lighthouse)

Comparison of Novatech, Amazon, and PWr WIT websites.

Table: Lighthouse performance metrics & page sizes

Website	Size	FCP	LCP	Speed Index	CLS
Novatech	6.1 MB	1.1 s	2.0 s	2.2 s	0.003
Amazon	22.9 MB	3.4 s	4.5 s	5.0 s	0.014
WIT Main	7.3 MB	2.7 s	2.7 s	2.7 s	0.004
WIT News	3.6 MB	2.7 s	2.7 s	4.1 s	0.022

Legend:

FCP (First Contentful Paint): Time until the first visible content appears.

LCP (Largest Contentful Paint): Time until the largest visible element is fully loaded.

CLS (Cumulative Layout Shift): Measures unexpected visual shifts (lower is better).

Key Takeaway

Page size alone does not dictate performance. Novatech achieved the best load times and layout stability despite being heavier than the WIT News page.

Automating the entire shopping workflow.

Implementation:
Generated using
Playwright Codegen to
mimic exact user
behavior.

Result: 100% pass rate without manual interaction.





Summary

- **Network Analysis:** AJAX updates ensure dynamic UI modifications without full page reloads, improving user experience.
- **Security:** Backend validation successfully prevents client-side DOM manipulation attacks (e.g., price tampering).
- **Performance:** Optimization of scripts and site structure is more crucial for load speeds than raw page size (Novatech vs. Amazon).
- **Automation:** Modern E2E tools like Playwright allow for reliable, automated validation of complex purchasing workflows.



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THE END